VERSION 0.4.0 · 18 OCTOBER 2026

Lumen

A native Windows shell around WebView2, written in Rust, built to keep memory low.

What it is so far, what it measures, and where it goes next.

19.2 MB
idle, Lumen itself

91 MB
one Wikipedia tab

1.1 GB
forty heavy tabs, asleep

47
tools for AI agents

Measured on one laptop, release builds, private working set. Lumen compared with earlier versions of itself; no comparison with other programs yet.

What it is

Lumen is a Windows program that shows web pages. It does not draw them. Pages are drawn by Microsoft WebView2, the Chromium-based engine already installed on every Windows 11 machine. Everything around the page is Lumen's own code: the window, the tabs, the sidebar, the content blocker, the downloads and the memory policy. That part is about 40,000 lines of Rust.

A page in Lumen costs about what the same page costs in any Chromium-based program, because it is the same engine. What Lumen controls is the rest: how heavy the shell is, how many renderer processes run, and what happens to pages you are not looking at. That is where the numbers below come from.

It is also a place for AI agents to work. Agents drive it over MCP, each in tabs and a profile of its own.

Spec

PART	DETAIL
Language	Rust 1.98, edition 2024
Runs on	Windows 10 and 11, x64
Page engine	Microsoft WebView2 (Chromium), through wry 0.57 and webview2-com 0.39
Window and interface	tao 0.37; every pixel drawn natively with Direct2D and DirectWrite, no HTML
Code	about 40,600 lines of Rust, one process for the shell
Binaries	`lumen.exe` 15.5 MB, `lumen-agent.exe` 2.0 MB (release, stripped)
Tests	337 unit tests, plus live suites that drive a real headless Lumen
Content blocker	adblock-rust 0.13, EasyList, EasyPrivacy, uBlock lists; about 1.7 µs a request
Agent interface	MCP: an HTTP endpoint on localhost behind a token, and a stdio bridge; 47 tools
Scripts	Luau (sandboxed, through mlua), for scripts the person writes and agents run
Data	`%APPDATA%\Lumen`: settings, session, filters, history, the WebView2 profile

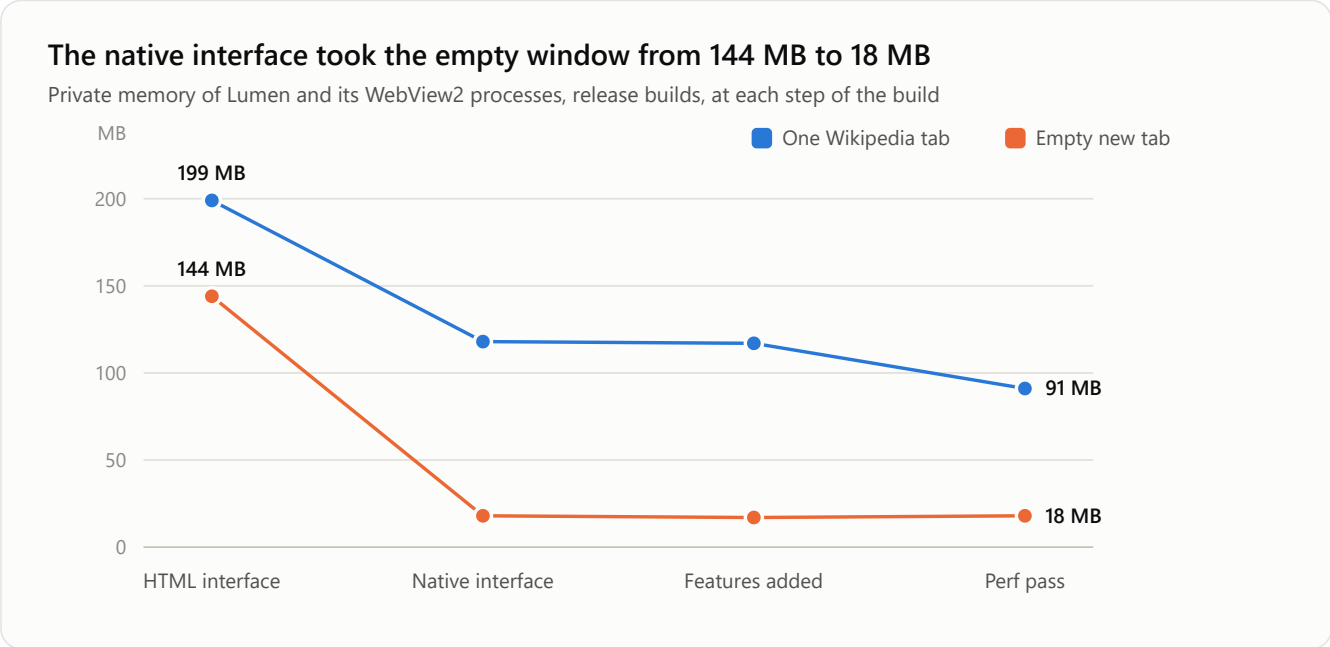
What it is built from, all pinned, anything that runs checked by SHA-256 before it runs:

PROJECT	WHAT IT DOES IN LUMEN	LICENSE
adblock-rust (Brave)	Blocks ads and trackers; hides leftovers with CSS	MPL-2.0
uBlock Origin 1.75.0 scriptlets	Strips ads out of YouTube's player data before the page reads it	GPL-3.0
EasyList, EasyPrivacy, uBlock lists	The rules: ads, trackers, cookie banners, tracking parameters	CC BY-SA / GPL
URLhaus and phishing filter lists	Malware and phishing hosts, refreshed weekly	MIT
Dark Reader 4.9.133	Dark pages for sites without a dark mode	MIT
Mozilla Readability 0.6.0, DOMPurify	Reader view, and article text for agents	Apache-2.0 / MPL-2.0
Bergamot	Page translation on the PC	MPL-2.0
aria2 1.37.0	Downloads with up to 16 connections, torrents	GPL-2.0
yt-dlp, FFmpeg	Saving the video on a page	Unlicense / GPL
wry, tao, windows-rs	The window, the WebView2 binding, every Win32 call	MIT / Apache-2.0

Performance

All numbers are measurements on one laptop: release builds, a throwaway profile, and Task Manager's "Memory" figure (private working set) unless a chart says otherwise. They compare Lumen with earlier versions of itself. A side-by-side run against other programs on the same pages has not been done yet, so there is no such claim here.

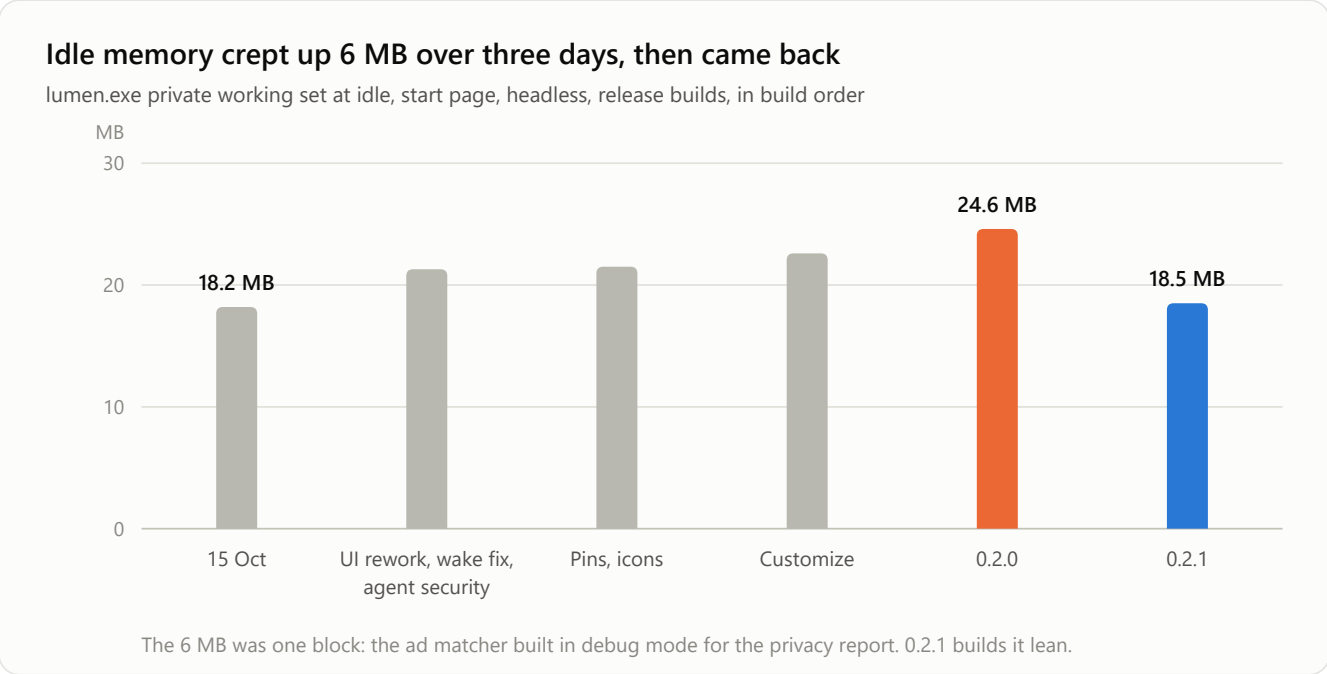
The shell



Replacing an interface written in HTML with one drawn natively took an empty window from 144 MB in five processes to 18 MB in one. A later pass on the event loop and on sleeping tabs took one Wikipedia tab from 117 MB to 91 MB, and its CPU from 0.47% to 0.16%.

STEP	EMPTY NEW TAB	ONE WIKIPEDIA TAB
HTML interface	144 MB	199 MB
Native interface	18 MB	118 MB
Features added	17 MB	117 MB
Perf pass	18 MB	91 MB

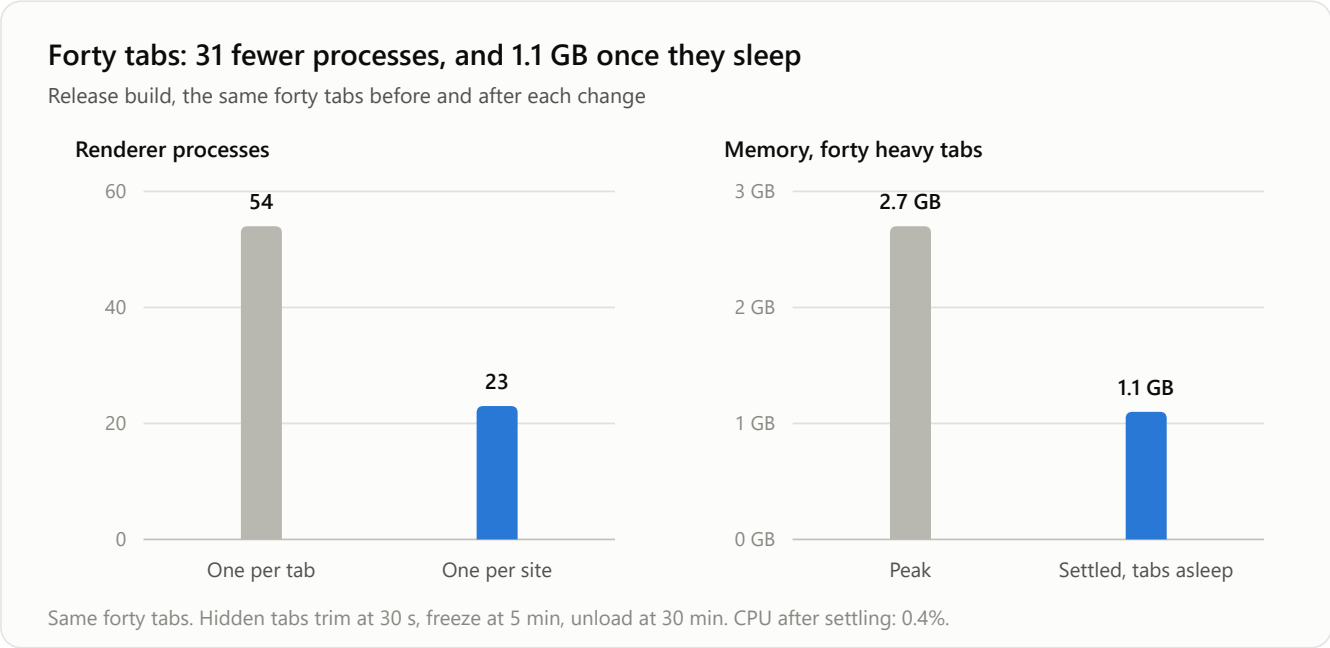
Idle



Idle memory crept up over three days of features. A count of the heap found almost all of the growth in one allocation: the content blocker's compiled rules, built in a debug mode that keeps every rule's text, only to tell an ad from a tracker in a report. 0.2.1 builds the rules lean and answers that question with a 380 KB table of tracker host hashes. The malware and phishing lists added in the same release hold 81,000 hosts the same way, for 0.5 MB.

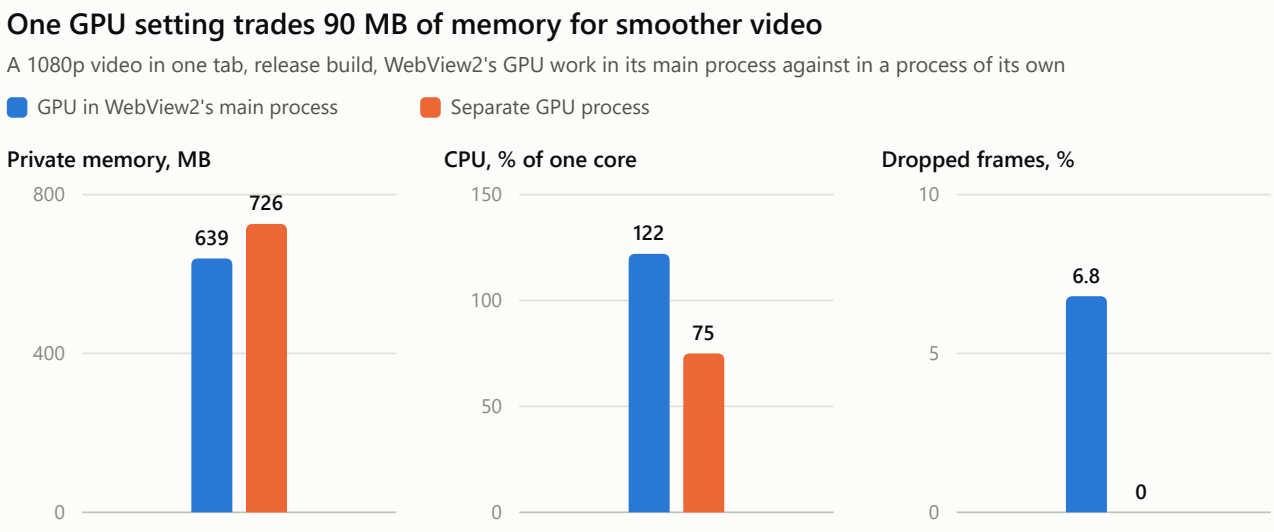
BUILD	IDLE
15 October	18.2 MB
UI rework, wake fix, agent security	21.3 MB
Pins, icons	21.5 MB
Customize	22.6 MB
0.2.0	24.6 MB
0.2.1	18.5 MB
0.4.0	19.2 MB

Many tabs



Two policies do this. One renderer per site instead of one per tab: forty tabs went from 54 renderer processes to 23, about 310 MB less. And tabs you are not looking at sleep: their caches are trimmed after 30 seconds, they freeze after 5 minutes and unload after 30. Forty heavy tabs peaked at 2.7 GB and settled at 1.1 GB, at 0.4% CPU.

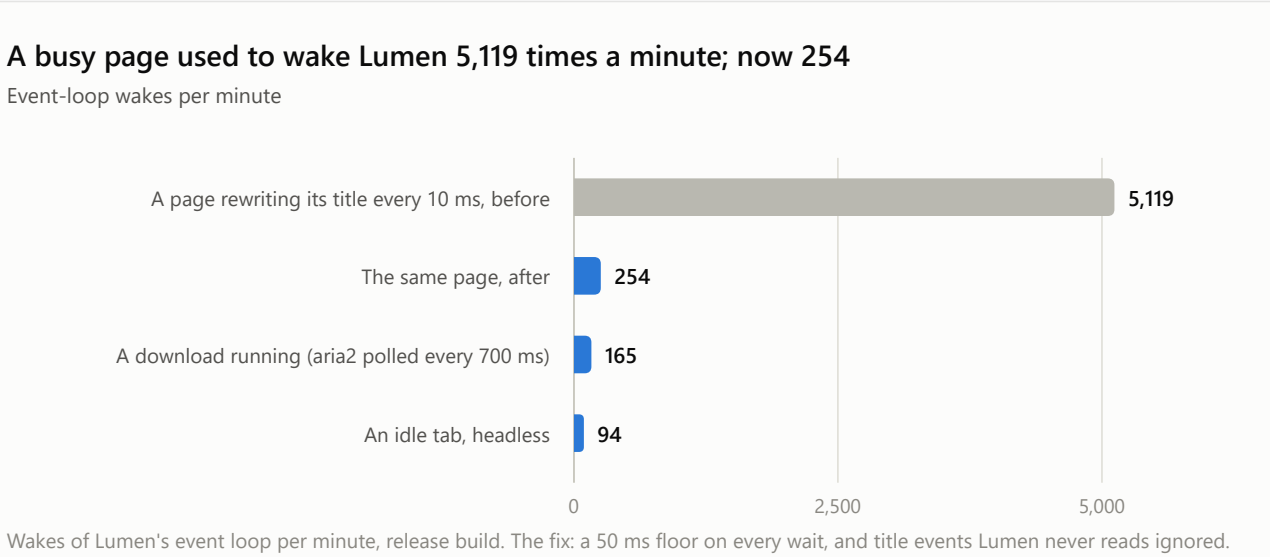
Video, and one trade-off



By default WebView2's GPU work runs inside WebView2's main process: one process and about 90 MB fewer. The cost shows in video: a 1080p video dropped 6 to 8% of its frames and used more CPU. A setting moves the GPU work back to its own process when smooth video matters more than memory.

	GPU IN WEBVIEW2'S MAIN PROCESS	SEPARATE GPU PROCESS
Private memory	639 MB	720 to 733 MB
CPU, one core	118 to 127%	70 to 80%
Dropped frames	5.8 to 7.8%	0%
Processes	8	10

Waking up



A page that rewrote its title every 10 ms woke Lumen's event loop 8,531 times in 100 seconds. Now the loop never waits for less than 50 ms, and it ignores events it never reads.

Other measured numbers

WHAT	NUMBER
One agent tab	about 85 MB
Agent tool call, a round trip to the window	0.7 ms (35 ms before 0.4.0)
Agent: fill and send a form, ten-step workflow	1.7 s and 2.6 s (5.7 s and 8.8 s before 0.4.0)
Agent: the ten-step workflow as a Luau script	1 call and 580 bytes back, against 12 calls and 4 KB
Content blocker, per request	about 1.7 µs
Usage journal, on against off	under 0.1% CPU, under 1 MB
CPU cap set to 50%	52% measured
Network cap set to 2048 KB/s	about 1,990 KB/s measured
A 100 MB download, before and after tuning aria2	17.8 s to 15.5 s mean (runs of the same build swung by 46%)
Servo, an experimental engine behind a build flag	renders Hacker News in 38 MB

What it does

- Sidebar or tab strip, in vertical, horizontal or hybrid layouts. Colours, fonts, shapes, motion, clock and wallpaper can all be changed.
- Content blocking for ads, trackers and cookie banners, on by default.
- Pop-ups blocked unless you clicked for them. Malware and phishing sites get a warning page.
- A privacy report: what was blocked and what was counted, with trends.
- Reader view, dark pages, translation on the PC, read aloud.
- Downloads through aria2, torrents, saving the video on a page.
- History you can search by what a page said.
- Pins, bookmarks with folders and import, a speed dial, workspaces, split view, tab search.
- KeePass logins filled from a `.kdbx` file, kept in memory only, locked after 10 minutes.
- Can be set as Windows' default for web links.
- A usage journal on the PC (memory, CPU, slow sites, crashes) that sends nothing anywhere.

For AI agents

Any agent on the machine can drive Lumen through MCP: Claude Code, opencode, Codex, Cursor. One program, `lumen-agent`, connects them. It finds Lumen and its token by itself, so an agent's config holds a path, never a secret.

- An agent sees a page as a snapshot: every heading and control, each with a short ref like `e5`. It clicks and types by ref, with real input events.
- Every agent gets its own tabs and its own profile, with its own cookies and storage. One agent cannot see another's tabs, and none can see yours unless you lend one.
- When a login, a captcha or a payment needs a person, the agent hands the tab over and waits.
- `lumen.exe --headless` runs Lumen off-screen: no focus, no taskbar button, no sound.
- Agent tabs live on a stage: a window of their own, off every screen, where Chromium still counts them as visible. Clicks land, and nothing comes to the front. Click an agent's tab to watch it live; the agent keeps working.
- Replies say what happened: a failed page's HTTP status, what a click changed or that it changed nothing. Page text is fenced, so a page cannot pose as the person.
- Fewer calls: `browser_batch` (up to 20 calls in one), `browser_fill` (a whole form), `browser_table`, `browser_links`. And Luau scripts: the person puts `.luau` files in a folder, agents list and run them. Every call a script makes is checked as if the agent made it.

The first real test, on 18 October, found reading fast and clicking broken: in a background tab a click did nothing and still answered "clicked". 0.3.0 fixed that with the stage, 0.4.0 made every call faster.

The design for that work is in the agent-side spec.

The plan

In order. Each step ships as its own version, with a line in the changelog.

1. **Honest replies and a smoke test.** A failed page says its HTTP status. A click says what it changed, or that nothing changed. Page text is marked as page text, so a page cannot pose as the person. Before every release, a live test clicks real buttons. **Done in 0.3.0.**
2. **The agent stage.** Agent tabs live in a window of their own, off every screen. WebView2 sees them as visible, so clicks land, and nothing comes to the front. **Done in 0.3.0.**
3. **Watching.** Click an agent's tab to see its live page, with the agent's own cursor in the accent colour. Your mouse cannot knock it off. **Watching is in 0.3.0; the cursor is next.**
4. **Talking to an agent.** A text field to steer an agent now, queue its next task, or start a new one.
5. **Speed for agents.** Fewer calls and smaller replies: batches of steps, Luau scripts, and higher-level tools for forms and tables. **Done in 0.4.0.** Snapshots that send only what changed are still to do.
6. **Headless trims.** Measure first, then cut what an off-screen Lumen still does for nobody.
7. **The comparison.** Lumen against other Chromium-based programs on the same pages and the same machine, published with the method.
8. **Source.** The code is meant to be published. Its license follows from what it bundles: uBlock's scriptlets are GPL-3.0.
9. **An installer.** Signed, so Windows does not warn about an unknown publisher, with updates that arrive by themselves.

Later, and only if they earn their memory: Servo for light sites, a local AI model reading the page through Ollama, and an optional PDF editor that starts only when asked.

Limits

- Windows only. WebView2 is the engine.
- No Chrome or Edge extensions. Ad blocking, dark pages, passwords through KeePass and translation are built in. Anything else is not there.
- Early. Built in five days by one person and Claude, with AI workers writing much of the code and every change reviewed before it landed. Expect rough edges.

The rule does not change: if a feature costs memory while you are not using it, it had better be worth it.